

A solution study of complex formation between iron(III) and oxalate in dimethylsulphoxide

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Summary

The complex formation between iron(III) and oxalic acid (ethanedioic acid, H₂ox) has been studied by potentiometry in dimethylsulphoxide (dmso) solution. H₂ox behaves as a weak diprotic acid in such a solvent, with overall association constants: log β_{j1} = 8.551(3) and log β_{j2} = 14.242(3) at 25°C and 0.1 M *n*-Bu₄NClO₄. A reliable set of overall stability constants for the iron(III)–oxalato complexes, log β_{11} = 13.16(4), log β_{12} = 23.66(4) and log β_{13} = 30.75(4), have been obtained for the first time under identical conditions. The electrochemical behaviour of such complexes was studied in dmso at a platinum electrode. The coordination ability of oxalate towards iron(III) in dmso and water media is compared and discussed in the light of thermodynamic and structural parameters.

Introduction

Water is by far the most widely used solvent in solution chemistry and most complex formation studies have been carried out in this medium. However, in certain cases water cannot be used as solvent due to: (i) hydrolytic reactions of highly charged metal ions; (ii) insolubility of the ligand or of one or more of the complexes to be formed; (iii) unfavourable acid–base ligand behaviour, for instance when the ligand is so weak as acid than it does not deprotonate in water, or when it is so strong than it is rarely protonated in this solvent.

Use of dimethylsulphoxide (dmso) or dmso/water mixtures as solvent avoids such difficulties in many cases^(1–10). Dmso is an aprotic polar solvent which readily dissolves many water-soluble salts as well as other sparingly soluble substances⁽¹¹⁾. Hydrolytic reactions of metal ions in dmso are inhibited in the absence of free hydroxide ion as shown recently⁽⁵⁾, even in the presence of significant amounts of water⁽⁶⁾. On the other hand, pure dmso as well as dmso/water mixtures are very appropriate solvents for potentiometric studies because of the Nernstian response of a combined glass electrode⁽³⁾. Such solvents provide a wider pH range than water, allowing deprotonation of very weak acids, which do not deprotonate in aqueous solution, to be studied. Generally, neutral protonic ligands exhibit smaller acidity constants in dmso than in water⁽¹²⁾, an observation especially relevant in solution studies of complex formation involving ligands such as oxalic acid (H₂ox). The obvious difficulties associated with determining accurate acidity constants in water⁽¹³⁾ as well as the hydrolysis of Fe³⁺ are non-existent in dmso, as demonstrated in this paper.

Keeping in mind the above outlined considerations, we have studied complex formation between iron(III) and

oxalic acid in dmso by potentiometry. The stability constants of the 1:1, 1:2 and 1:3 (metal:ligand) complexes, as well as the H₂ox acidity constants, have been determined. An electrochemical study of such species has also been carried out.

Results and discussion

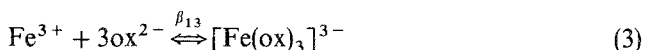
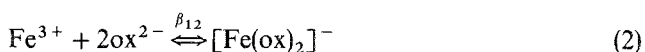
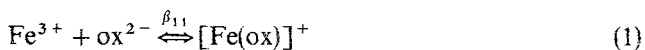
Oxalic acid acidity constants

Oxalic acid is a diprotic acid which behaves as a strong acid for the first proton dissociation and moderately strong for the second in aqueous solution⁽¹³⁾. Accurate acidity constants of oxalic acid are needed in order to determine reliable stability constants of oxalato complexes because deprotonation equilibria and complex formation occur in the same pH range. Nevertheless the determination is not easy with H₂ox, as shown by the range of pK_{a1} values reported in the literature.

In contrast to water, dmso is an ideal solvent for studying deprotonation equilibria of H₂ox and, consequently, its complex formation with metal ions. Dmso solutions of oxalic acid (10^{−2} M) were titrated with 0.1 M tetra-*n*-butylammonium hydroxide and the e.m.f. data were processed by the SUPERQUAD program⁽¹⁴⁾ allowing us to determine reliable proton association constants, 8.551(3) and 14.242(3) for log β_{j1} and log β_{j2} , respectively (maximum standard deviation in the last significant digit is given in parentheses). Throughout the data treatment, 100 experimental points from two experiments were used covering the 3.6–8.4 pH range. The distribution diagram for H₂ox as a function of pH is depicted in Figure 1. Oxalic acid is the main species at lower pH values; Hox[−] is the predominant one in the narrow 6–8 pH range. Oxalate dianion is the only existing species at pH values > 10. As far as this species is concerned we observed only up to a 40% degree of formation because of the unsatisfactory response of the electrode in such a basic medium.

Stability constants of iron(III)–oxalate

A series of dmso solutions of iron(III) nitrate and oxalic acid ($C_M = 1.0\text{--}5.0 \times 10^{-3}$ M, $C_L = 1.0\text{--}15 \times 10^{-3}$ M) were titrated with tetra-*n*-butylammonium hydroxide in order to investigate the formation of oxalato complexes. Data processing by SUPERQUAD of 120 experimental points from different experiments, carried out in the 2.0–6.0 pH range, provide us with the stability constants for the equilibria; [Equations (1)–(3)]:



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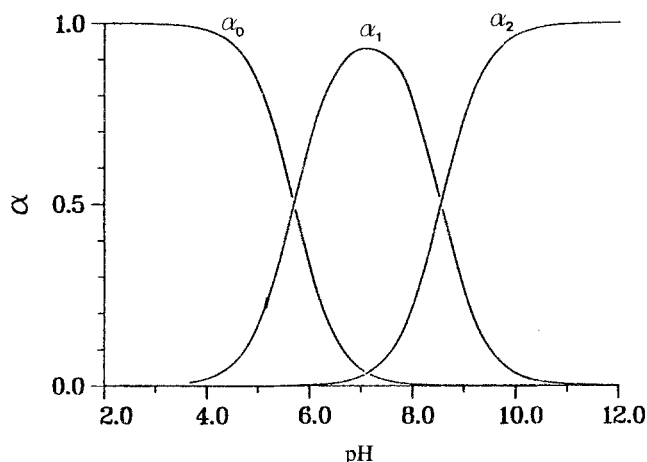


Figure 1. Distribution diagram (α versus pH) for oxalic acid: $\alpha_0 = [\text{H}_2\text{Ox}]$, $\alpha_1 = [\text{Hox}^-]$ and $\alpha_2 = [\text{Ox}^{2-}]$.

$\log \beta_{11} = 13.66(4)$, $\log \beta_{12} = 23.66(4)$ and $\log \beta_{13} = 30.75(4)$. Several models involving hydroxo and hydrogenoxalato species as well as oxalato-bridged iron(III) complexes were ruled out by the SUPERQUAD program in the process of refinement and only the above-mentioned species fitted well the experimental data in the pH range investigated. The distribution diagram, α versus pH, for the existing species in solution is shown in Figure 2. The formation of 1:1, 1:2 and 1:3 (metal:ligand) complexes occurs at $\text{pH} < 6$, i.e. in the pH range where oxalate is present mainly as H_2Ox .

The fact that iron(III) hydrolysis was ruled out through the processing data is easily understood because the constant for the hydrolysis [Equation (4)]



in a dmsO:water (80:20) mixture is $\log K_h = 4.52(6)$, ca. 100 times smaller than the corresponding value in water⁽¹⁵⁾. This value must be even smaller in dmsO with lower water content. Moreover, there is no free iron(III) in dmsO at $\text{pH} \geq 2.5$ in the presence of oxalate, as shown in Figure 2. On the other hand, although iron(III) complexes where oxalate behaves as a bridged bis-bidentate ligand are known in the solid state^(16–18), their formation in solution has not been observed as far as we are aware. This is probably because concentrated solutions are required for such polynuclear species to be formed, but the usual spectrophotometric and potentiometric techniques are not operative at such concentration levels. Finally, although examples of coordination of Hox^- are known, both in solution⁽¹⁰⁾ and in the solid state⁽¹⁹⁾, this coordination mode is rarer and has not been observed in the present study under our experimental conditions. At this content, although Bauer and Smith have determined the stability constant of the protonated monooxalato species, $\text{Fe}(\text{HC}_2\text{O}_4)^{2+}$, in aqueous solution⁽²⁰⁾, its existence in this medium has been questioned by other authors^(21, 22).

The high values of the stability constants for 1:1, 1:2 and 1:3 iron(III):oxalate complexes in dmsO are consistent with the chelating character of the oxalate ligand as illustrated by the reported structures of oxalate-containing iron(III) species^(18a, 22). These values

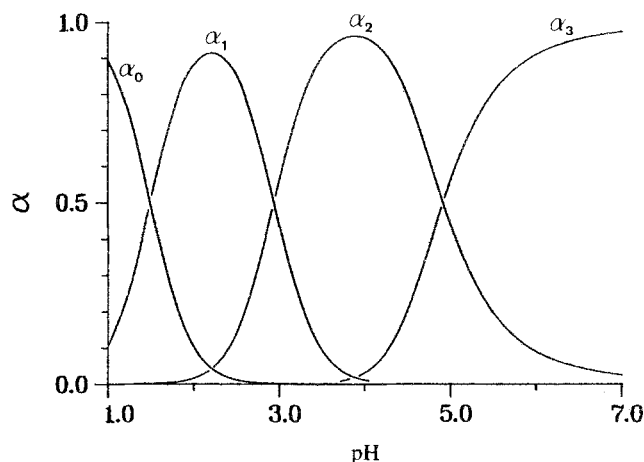


Figure 2. Distribution diagram (α versus pH) for iron(III)-oxalate system: $\alpha_0 = [\text{Fe}^{3+}]$, $\alpha_1 = [\text{Fe}(\text{ox})^+]$, $\alpha_2 = [\text{Fe}(\text{ox})_2^-]$ and $\alpha_3 = [\text{Fe}(\text{ox})_3^{3-}]$.

are greater than the corresponding ones for such species in aqueous solution⁽¹³⁾. To the best of our knowledge, it is the first time that the complex formation between iron(III) and oxalate has been investigated over a wide pH range. The properties of solvent dmsO outlined above have made possible such a complete study. Previous work on this system was carried out in aqueous solution at very low pH values. Under these conditions, potentiometric measurements are less precise and deprotonation of oxalic acid and complex formation occur simultaneously, complicating the study because of the inaccuracy of the first dissociation constant of the acid. All these problems are overcome in dmsO as shown in this paper.

Electrochemical study

We have studied, by cyclic voltammetry, the system $\text{Fe}^{3+}/\text{ox}^{2-}$ in dmsO.

The 1:1 iron(III):oxalate species show a totally irreversible one-electron reduction peak A_2 at 0.03 V [Figure 3(b)] (s.c.e. electrode). However, an oxidation peak A'_1 at 0.30 V is detected in the reverse sweep, corresponding to the $\text{Fe}^{2+}/\text{Fe}^{3+}$ couple as confirmed by comparison with the redox behaviour of a sample of Fe^{3+} under the same experimental conditions [Figure 3(a)]. From these results we infer that a fast dissociation of the coordinated oxalate lies at the heart of the observed irreversibility.

The cyclic voltammogram for a 1:2 metal to ligand molar ratio exhibits a totally irreversible reduction peak A_3 at ca. -0.4 V [Figure 3(c)]. In the reverse sweep, only the oxidation peak A'_1 is observed. The A_1 peak is not detected in the cathodic sweep after scanning the A'_1 one; this is probably due to recombination of Fe^{3+} with the free oxalate ligand.

The reduction potential is shifted to higher cathodic potential values for 1:3 or greater molar ratios. A totally irreversible reduction peak A_4 is now detected at -0.55 V [Figure 3(d)]. Only the A'_1 peak is observed in the reverse sweep; however, the oxidation peak A'_2 can be detected as a reversible couple (A_2/A'_2) at sweep rates greater than 0.55 V s^{-1} . All these results allow us to propose the following reactivity pattern:

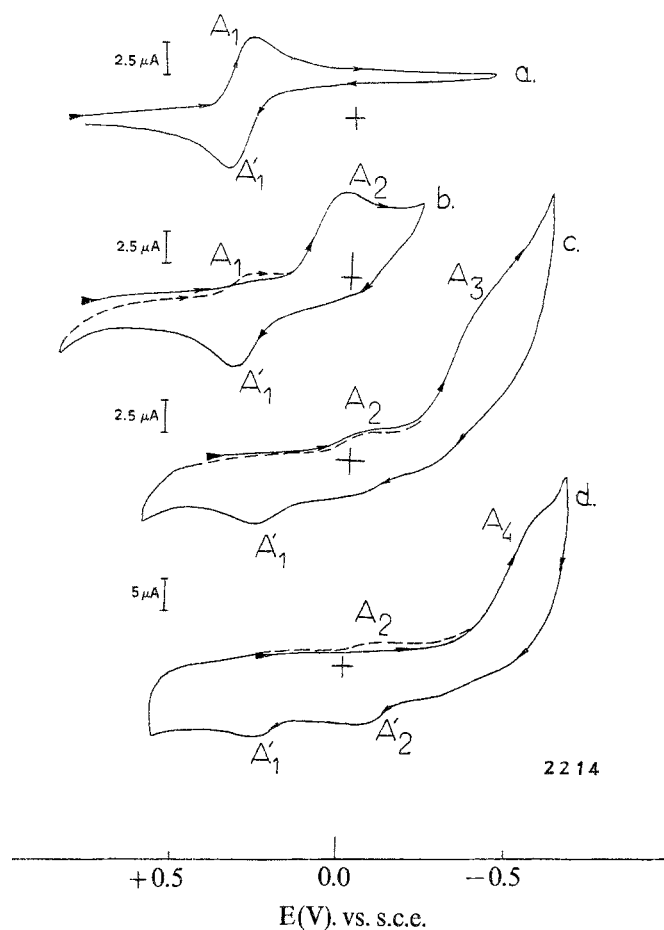
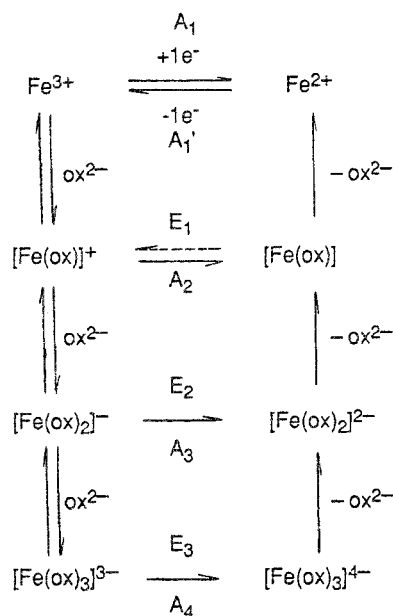


Figure 3. Cyclic voltammograms for (a) $[\text{Fe}]^{3+}$ (b) $[\text{Fe}(\text{ox})]^{+}$ (c) $[\text{Fe}(\text{ox})_2]^{-}$ and (d) $[\text{Fe}(\text{ox})_3]^{3-}$. Scan rates 100 mV s^{-1} (a, b and c) and 550 mV s^{-1} (d).



Scheme 1.

It is well known that aqueous solutions of iron(III)–oxalate are unstable because of the photosensitivity of tris(oxalate)ferrate(III) which generates insoluble ferrous oxalate⁽²³⁾. However, dmso solutions of this system are stable over several weeks. In order to explain the higher stability of the Fe^{3+} –oxalate system in dmso *versus* H_2O we have also made cyclic voltammograms of separated

aqueous solutions of Fe^{3+} and oxalic acid with 0.1 M HClO_4 as supporting electrolyte in order to avoid iron hydrolysis. The $\text{Fe}^{3+}/\text{Fe}^{2+}$ system is observed as a typically reversible one-electron redox couple ($E_p^{\text{red}} = +0.55 \text{ V}$). This value contrasts with the corresponding one in dmso ($E_p^{\text{red}} = +0.30 \text{ V}$) and provides evidence for the higher oxidizing power of Fe^{3+} in water. Additionally, the oxidation of oxalate in water occurs more easily than in dmso. Consequently, the redox reaction between Fe^{3+} and oxalate is less favoured thermodynamically in dmso than in water.

Experimental

Reagents

$\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$, $\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ and dmso (analytical grade reagents) were used without further purification. $n\text{-Bu}_4\text{NClO}_4$ and piperidinium oxalate dihydrate were obtained and purified as previously described^(10,17). Iron(III) solutions were standardized by atomic absorption spectroscopy. Carbonate-free solutions of $n\text{-Bu}_4\text{NOH}$ in dmso (freshly prepared from BDH reagent; 40% aqueous solution) were used as titrant in the potentiometric study. 0.1 M $n\text{-Bu}_4\text{NClO}_4$ was used as background electrolyte in potentiometric and electrochemical experiences in dmso.

Physical measurements

Potentiometric titrations were carried out using a previously described titration assembly⁽¹⁰⁾. The temperature of all solutions was maintained at $25 \pm 0.1^\circ \text{C}$ by circulating constant-temperature water through the water-jacketed titration cell. Nitrogen presaturated with dmso was passed over the surface of the solution. Solutions were kept in the dark during the titration process. Nernst's equation was strictly obeyed over the hydrogen ion concentration range investigated. The electrode was calibrated as a $[\text{H}^+]$ probe, but as a matter of convenience, pH instead of $-\log[\text{H}^+]$ has been used throughout the text. Accurate concentrations of the basic titrant as well as of the E^0 value were determined with toluene-4-sulphonic acid solution in dmso before each titration series by applying the Gran method. Reliable potential measurements were obtained within 5 and 10 min after each addition except in the more basic region where more time is required to get stable lectures. The computer program SUPERQUAD was used to process e.m.f. data from each experiment and to calculate the stability constants of the interaction between iron(III) and oxalate.

The electrochemical measurements were carried out in a three-electrode cell under an Ar atmosphere. The working and auxiliary electrodes were platinum, the reference electrode was a saturated calomel electrode, electrically connected to the dmso solution by a salt bridge containing the non-aqueous solvent and the supporting electrolyte. Cyclic voltammograms were obtained with a 305 H.Q. Instruments programming function generator which was connected to a 552 Amel potentiostat, and recorded with a Riken–Denshi F35 x–y recorder. The system was calibrated against cobaltocene, $[\text{Co}(\eta = \text{Cp})_2]$.

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